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Assistant Professor

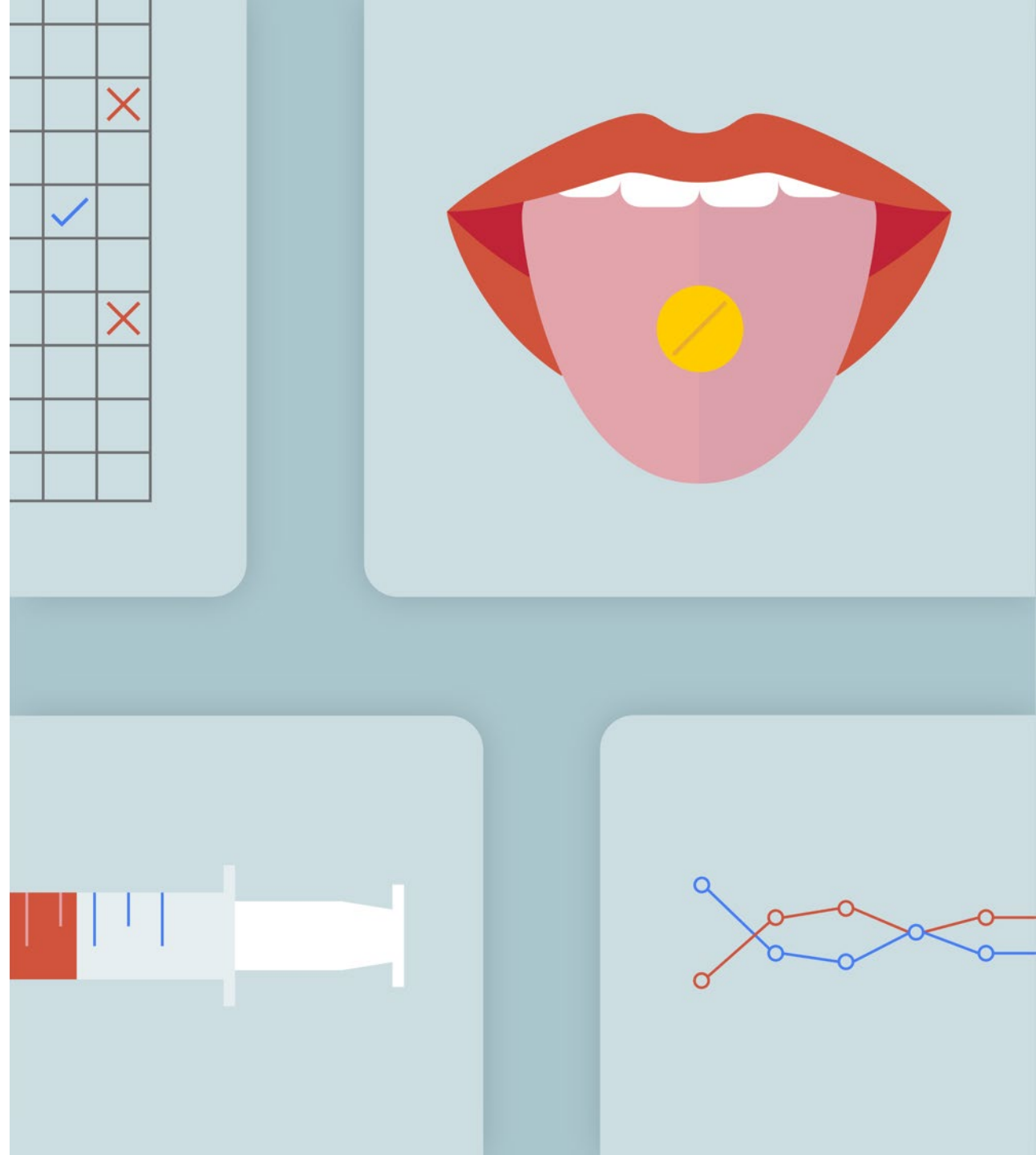
R crash course

session 3



Utrecht University

Pharmacoepidemiology
and Clinical Pharmacology





Preparation for session 3

walkthrough



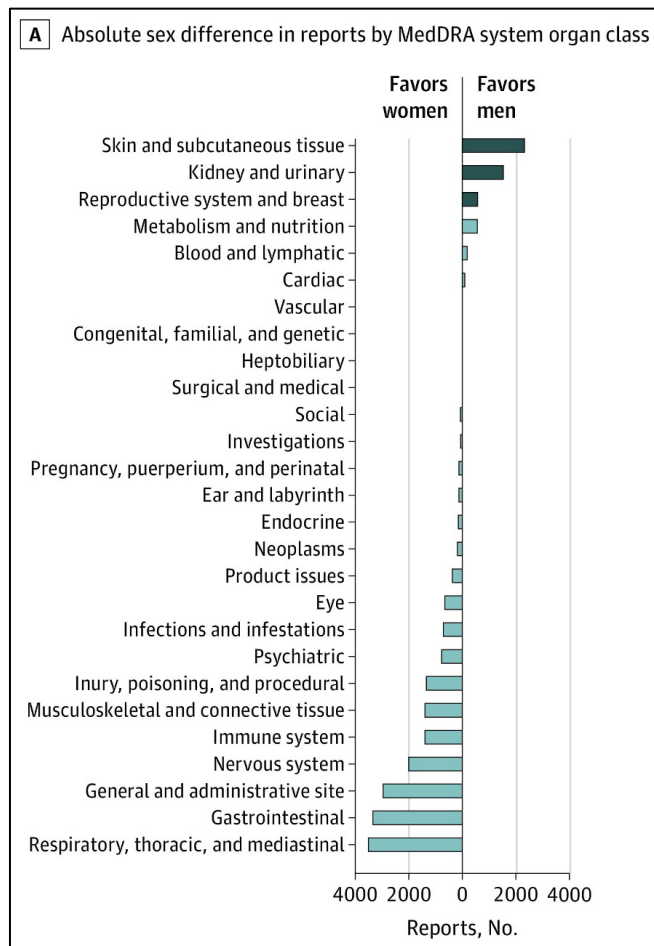
Descriptive data analyses

Part 1
learning goal 6



Two flavours

Graphs



Tables

Table. Most Commonly Reported ADRs by Sex^a

ADR rank by frequency of reporting	ADR type (No. of reports)			
	VigiBase		Lareb	
	Women	Men	Women	Men
1	Cough (10 909)	Cough (7701)	Angioedema (199)	Angioedema (153)
2	Angioedema (3441)	Angioedema (6634)	Cough (163)	Cough (124)
3	Dizziness (2509)	Acute kidney injury (2830)	Therapeutic response unexpected (70)	Therapeutic response unexpected (91)
4	Drug hypersensitivity (2323)	Hyperkalemia (2159)	Dizziness (51)	Dizziness (49)
5	Headache (1965)	Dizziness (2056)	Headache (51)	Pruritus (45)
6	Nausea (1810)	Hypotension (1949)	Alopecia (44)	Erectile dysfunction (37)
7	Acute kidney injury (1793)	Dyspnea (1540)	Dyspnea (42)	Fatigue (36)
8	Dyspnea (1723)	Pruritus (1309)	Nausea (39)	Myalgia (29)
9	Drug ineffective (1605)	Drug ineffective (1272)	Paresthesia (37)	Headache (29)
10	Pruritus (1456)	Headache (1213)	Rash (35)	Muscle spasms (27)

Two flavours

Graphs

Tables



tableone package

Creating baseline tables with *tableone*

Baseline table describes your study population

```
CreateTableOne(data = dataset)
```

Several customisation options

- Choose variables
- Select variable types
- Select variable distribution (normal / non-normal)
- Strata



Statistical data analyses

Part 2
learning goal 6



What do you want to do?

Compare two sample means ==>

t-test

Compare two sample frequencies ==>

Chi-square

Correlate two continuous variables ==>

linear regression

Perform survival analysis ==>

Cox regression

What do you want to do?

Compare two sample means ==>

t-test

```
t.test(variable1 ~ variable2, dataset)
```

Additional options for

- Paired data
- One-sample t-test

What do you want to do?

Compare two sample frequencies ==> Chi-square

create dataframe that is the relevant 2x2 table

chisq.test(dataframe)

Function naming follows convention *[testname].test()*

What do you want to do?

Correlate two continuous variables ==> linear regression

Perform survival analysis ==> Cox regression

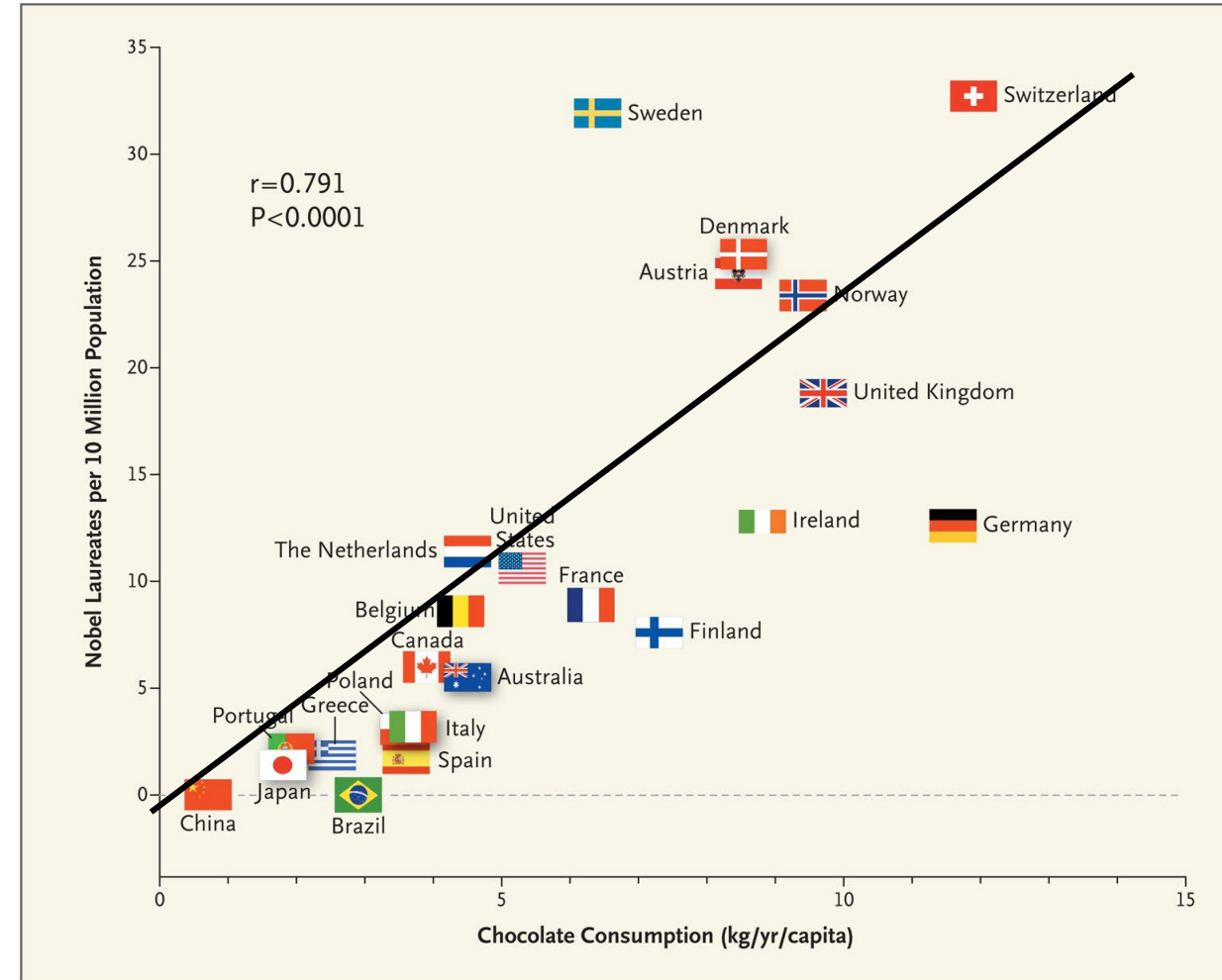
Regression basics

$$y = ax + b + \textit{error}$$

Regression – the basics

$$y = ax + b + \text{error}$$

$$N_{\text{nobel}} = \textcircled{a} * \text{choc_consumption} + \text{y_intercept} + \text{error}$$



Linear regression in R

$$y = ax + b + \text{error}$$

stats package (automatically loaded) → *lm()* function

```
> summary(lm(data = dat_long, hb ~ bmi))
```

Call:
lm(formula = hb ~ bmi, data = dat_long)

Residuals:

Min	1Q	Median	3Q	Max
-2.6335	-0.5590	-0.0415	0.5176	4.6400

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	7.911272	0.0100398	787.98	<2e-16 ***
bmi	0.0239703	0.0003148	76.14	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8105 on 370191 degrees of freedom
Multiple R-squared: 0.01542, Adjusted R-squared: 0.01542
F-statistic: 5797 on 1 and 370191 DF, p-value: < 2.2e-16

Cox regression in R

survival package (must be loaded) → *coxph()* function

```
coxph(formula = Surv(fu_time, outcome_status) ~ indep.variable,  
       data=data)
```

Kaplan-Meier plot

```
plot(survfit(formula = Surv(time, status) ~ indep.variable,  
             data=data)
```

One step further than the basics

- adding confounders to your model

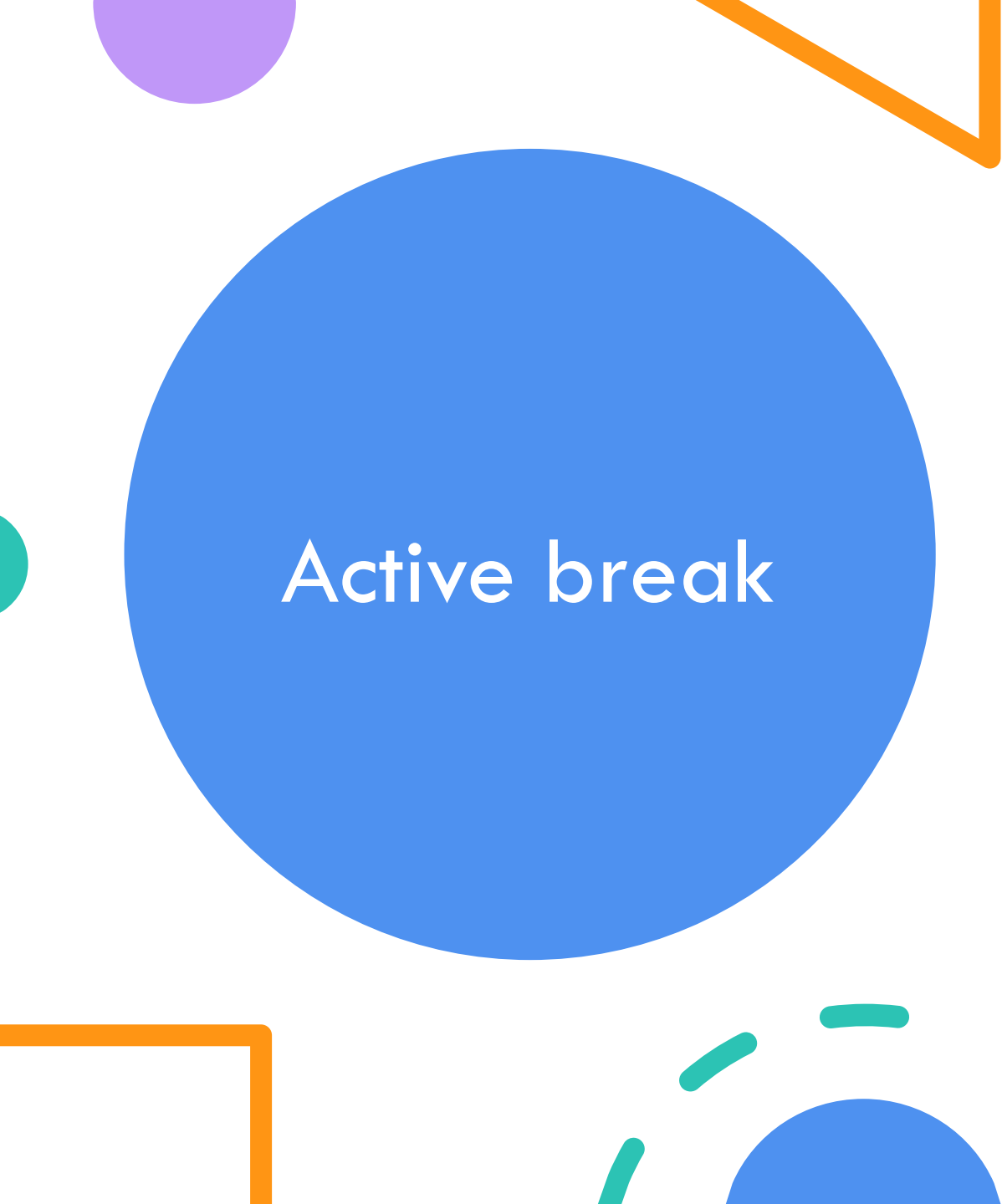
`lm(y ~ x + confounder1 + confounder2 ..., data)`

- getting the output in a nice format

save the model to an object

`model1 <- lm(y ~ x, data)`

`summary(model1)`



Active break

Run at least one statistical analysis

Guidance:

[Cookbook for R Chapter 7](#) (especially sections 2, 3, 5)

[Survival analysis with R](#)

[Survival package vignette](#)

Learning goals covered

1. Perform basic coding tasks in RStudio like creating variables, and indexing data frames
2. Apply good workflow practices for coding, scripting, and getting help
3. Explain the rationale behind the tidy data framework and why this benefits data analysis
4. Transform datasets with the *dplyr* package
5. Visually explore datasets with *ggplot2* and interpret the results (exploratory analysis)
- 6. Perform descriptive and statistical analyses and interpret the results**

Revise: check if you understood everything

Introduction to R for health data analysis

- Chapter 8

Cookbook for R

- Chapter 7, sections 7.1-7.5

See course outline for additional revision materials and other optional further reading

5 minute feedback form

Please share your thoughts on the course so we can improve it for next year!

<https://forms.gle/pfZqzW5n37BCUYow6>

The background is white and features several abstract geometric elements. In the top left, there are two short, vertical teal dashes. To their right is a large, solid purple circle. Further right, an orange line forms a right-angled corner, with one segment extending horizontally and the other vertically. In the top right corner, a portion of a teal circle is visible. On the left side, there is an orange square outline. Below it, a series of four teal dashes of varying lengths and orientations are arranged in a loose, curved pattern. A large, solid blue circle occupies the right half of the slide, serving as a backdrop for the text.

Good luck with your
project!